

S.I.G.S.M.

Inglés

NovaTech

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SEGUNDA ENTREGA

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ABSTRACT

Reasons:

Healthcare institutions face continuous challenges regarding operational costs and patient communication efficiency. At Hospital de Clínicas, paper-based document distribution generates substantial expenses and slows down administrative processes. To address these limitations, Novatech presents an integrated software system designed to modernize information access and streamline internal logistics.

Objectives:

The main objective of this project is to develop a web-based management system featuring instant QR-code document access for patients and an integrated ambulance traceability module for administrative staff.

Methodologies:

The system was designed using standard software engineering practices and developed in Visual Studio Code as the primary IDE (Integrated Development Environment), with Bootstrap for a responsive user interface and MySQL hosted on Rocky Linux 9.7 for secure database management and server stability.

Results:

The patient module provides immediate digital access to medical guidance and simple, user-friendly satisfaction surveys, eliminating dependence on printed materials. Additionally, the logistics module enables operators to monitor ambulance status and view essential transport data, improving overall data administration across the institution.

Conclusions:

Implementation of this platform significantly enhances patient experience and optimizes administrative workflow. Most notably, replacing paper documentation with digital solutions is estimated to save Hospital de Clínicas approximately 2.3 million Uruguayan pesos per year.

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Swot analysis

Strengths

- **Strong Team Communication:** Effective group dynamics with structured weekly meetings to keep project milestones on track.
- **User-Centric Design:** Intuitive, friendly, and accessible user interface tailored for both patients and healthcare personnel.
- **Robust Infrastructure Base:** High server stability, security, and performance guaranteed by Rocky Linux 9.7 deployment.
- **Effective Tech Stack Integration:** Seamless setup and full compatibility between MySQL database, Bootstrap framework, and Visual Studio Code environment.

Weaknesses

- **Limited Previous Framework Experience:** Initial learning curve regarding advanced features in certain web development tools and frameworks.
- **Parallel Module Workload:** High workload distribution required to balance two distinct system modules (patient documentation and vehicle tracking) within the development phase.

Opportunities

- **Institutional Alignment:** Strong willingness from Hospital de Clínicas to digitalize administrative processes and cut paper costs.
- **System Scalability:** High potential to adapt and expand this software architecture to other public healthcare institutions across Uruguay.
- **Environmental & Economic Impact:** Opportunity to generate substantial budget savings and support paperless eco-friendly initiatives.

Threats

- **Infrastructure Contingencies:** Potential network disruptions, hardware failures, or server downtime at the hospital's floor 6 DTI facility.
- **User Adaptation Barriers:** Possible resistance from administrative staff when transitioning from traditional manual workflows to the new digital platform.
- **Academic Time Constraints:** Rigorous deadlines across multiple course subjects that limit the buffer time for testing and revisions.

